Undergraduate Mathematics Symposium

Saturday, October 1, 2011
University of Illinois at Chicago

Schedule of Activities

8:15am – 8:55am	Sign-in and coffee in SEO 300
Morning Session – Plenary lectures, 50 minutes each, SEO 636	
8:55am	Opening remarks
9:00am	Yvonne Lai (U Michigan) **Decomposing Polyhedra**
10:00am	Dhruv Mubayi (UIC) **From Ramsey theory to arithmetic progressions and hypergraphs**
11:00am	Christopher Leininger (UIUC) **Geometry and dynamics of surface homeomorphisms**
Noon – 1:30pm	Lunch in SEO 300
Afternoon Session 1 – Student lectures, 20 minutes each, SEO 636	
1:30pm	Sean McAfee (UIC) **On improving Cayley's theorem for groups of order p^4**
2:00pm	Jonathan Gleason (U Chicago) **The F*-algebraic Formulation of Quantum Mechanics**
2:30pm	Christopher Perez (CalTech) **Siegel modular varieties: arithmetic invariants, cusps, and real points**
2:50pm – 3:30pm	Coffee break in SEO 300
Afternoon Session 2 – Student lectures, 20 minutes each, SEO 636	
3:30pm	Joshua Jay Herman (UIC) **Introduction to topological quantum computation**
4:00pm	Anita Thomas (IIT) **Sampling Within k-Means Algorithm to Cluster Large Datasets**
4:30pm	Jahan Claes (U Chicago) **Spectral Rigidity on Flat Tori**
5:00pm	In-Jee Jeong (Brown) **Bipartite Graphs, Quivers, and Cluster Variables**

Abstracts of Plenary Lectures

Yvonne Lai (U Michigan)

Decomposing Polyhedra

Suppose you are given two polygons of equal area. Can you slice up one polygon with scissors, rearrange the pieces, and obtain the other polygon? What if we ask the analogous problem of polyhedra? How we answer this question depends on how we define slicing and rearranging.

Starting from natural but different definitions leads to radically disparate answers. We will examine these questions from several perspectives, winding our way from Euclid's Elements to foundational mathematics of the 20th century.

Christopher Leininger (UIUC)

Geometry and dynamics of surface homeomorphisms

Surfaces are fundamental objects in mathematics arising in a number of different fields. Although compact surfaces were classified 150 years ago, they continue to be central to many areas of mathematics. In this talk, I'll give some indication of Thurston's classification theorem for surface homeomorphisms, primarily through examples. Then I'll focus on the "generic" type, the *pseudo-Anosov* homeomorphisms. I will explain some of the connections between geometry, topology and dynamics for this class of homeomorphisms.

Dhruv Mubayi (UIC)

From Ramsey theory to arithmetic progressions and hypergraphs

I will begin by giving a brief introduction to Ramsey theory, in layman's terms, and describing one of the most well-known open problems in this area. This will naturally lead to a similar question about arithmetic progressions in the integers. Finally, we will see a glimpse of how these problems about arithmetic progressions are solved by modern combinatorial methods using hypergraphs.